

Module 2: Data Preprocessing & Normalization

Binning, Chi-Square Correlation, Min-Max, Z-Score & PCA

Module II: Data Preprocessing & Normalization — Topics Covered

1. Data Cleaning & Smoothing by Binning
2. Correlation Analysis: Chi-Square (χ^2) & Pearson r
3. Min-Max Normalization & Z-Score Scaling
4. Dimensionality Reduction via PCA
5. Discretization & Concept Hierarchies

1. Data Transformation: Normalization Formulations

Normalization Method	Formula	Key Use Case
Min-Max Normalization	$v' = \frac{v - \min_A}{\max_A - \min_A} (\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A$	Maps values linearly into predetermined interval $[0, 1]$.
Z-Score (Zero-Mean)	$v' = \frac{v - \mu_A}{\sigma_A}$	Useful when min and max are unknown or outliers dominate.